Culminating Activity

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Abstract

Provide a concise summary of the study including purpose, methods, results, and conclusions (150–250 words).

Chapter 1: Introduction

Background of the Study

Global sales prediction plays an important role in helping businesses make informed decisions related to marketing strategies, inventory management, budgeting, and future business planning. Many organizations rely on predictive models such as Multiple Linear Regression to analyze trends and forecast sales performance using historical and market-related data. However, one of the common challenges in predictive analytics is the presence of missing data within datasets. Missing values may occur due to incomplete records, human error, system failures, or inconsistent data collection processes. When these missing values are not properly handled, they can negatively affect the accuracy, consistency, and reliability of sales prediction models, leading to poor decision-making and inaccurate forecasts.

This research focuses on examining the effect of missing value handling on improving global sales prediction using Multiple Linear Regression. Specifically, it aims to understand why missing data reduces model performance and how different data cleaning techniques can enhance predictive accuracy. It also highlights the significance of proper data preparation in machine learning and statistical analysis. By identifying effective preprocessing methods, the research can help businesses, researchers, and data analysts develop more reliable prediction models and improve the quality of data-driven decisions in global sales forecasting.

Statement of the Problem

1. What is the effect of missing value handling on improving global sales prediction using Multiple Linear Regression?
2. Why does missing data reduce the accuracy and reliability of sales prediction models?
3. How does applying data cleaning techniques improve the performance of the Multiple Linear Regression model?

Objectives of the Study

General Objective

- To investigate the effect of missing value handling on improving global sales prediction using Multiple Linear Regression.

Specific Objectives

- To determine how missing data affects the accuracy and reliability of sales prediction models.
- To implement and assess data imputation strategies to prevent the loss of valuable data caused by deleting incomplete rows.

Significance of the Study

This study mainly benefits stakeholders who utilize predictive analysis to guide data-driven decision making. By emphasizing the influence of missing data found within datasets used, as well as determining the impact it has to the model's efficacy, the following groups stand to gain:

- Business Organizations: Companies heavily rely on these predictive models to analyze trends and forecast sales performance. By proving how the influence of missing data skews the results of analysis, the marketing and sales departments of these organizations can refine and regulate their data collection procedures. These practices upholding data integrity can ensure quality regarding sales forecasting and, in turn, aid in their future executive business decisions and resource allocation.
- Data Analysts: This paper can serve as a benchmark to creating and optimizing new prediction models, especially addressing the challenges posed by missing values in their datasets. Analysts can utilize these findings to theorize more of these sophisticated strategies to create more robust, reliable, and statistically accurate analytical tools.
- Researchers: Practitioners and professionals under the field of data science and econometrics can refer to this paper. It offers a framework for future studies to compare different machine learning algorithms or imputation techniques surrounding the Multiple Linear Regression model used here.

Scope and Limitations

This research focuses on predicting global video game sales by comparing two data-handling methods which are deleting missing rows and using the Bayesian Linear Regression imputation. The analysis is specifically performed using Multiple Linear Regression within the R programming environment. The primary goal is to determine which data-handling technique leads to a more accurate and reliable prediction model.

Chapter 2: Review of Related Literature

Related Studies

In a study focused specifically on video game sales data from Kaggle, Affan et al. (2024) developed a machine learning-based sales prediction model using several regression algorithms. Their model made use of regression algorithms, including Linear Regression, Ridge Regression, and Gradient Boosting Regressor, to forecast worldwide video game sales based on variables such as genre, platform, publishers, critic reviews, and user ratings. Their work demonstrates that the same video game sales dataset used in this study has been applied in prior research, validating its academic relevance.

Separately, a study on predicting video game sales using supervised learning from the same Kaggle dataset highlighted the impact of missing data on model outcomes. After downloading the dataset, the researcher dropped rows with empty values or filled in missing values to standardize the columns before applying regression modeling, noting that the dataset contained missing score fields for some video games, which posed challenges to prediction reliability. This finding directly parallels the core concern of this research, that unhandled missing data undermines prediction accuracy. Du (2025).

Ahmad et al. (2024) conducted a comprehensive review of machine learning applications specifically for missing value imputation. Their analysis sought to enhance researchers' understanding of MVI methods and facilitate the development of robust interventions in data preprocessing for data analytics, elucidating the strengths and limitations of different ML algorithms in imputing missing values across different types of data. This reinforces the idea that the choice of imputation technique has measurable consequences for downstream model performance.

Related Literature

Multiple Linear Regression

Multiple Linear Regression is a fundamental statistical technique used to model the relationship between one dependent variable and two or more independent variables. According to Fiveable (2024), the goal of MLR is to find the line of best fit that minimizes the sum of squared residuals, where the model takes the form $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \epsilon$, with each slope coefficient representing the change in the dependent variable for a one-unit change in the corresponding independent variable, holding all others constant.

The reliability of MLR estimates depends on a set of classical assumptions known as the Gauss-Markov conditions. As explained by Jim (n.d.), the Gauss-Markov theorem states that if a linear regression model satisfies the classical assumptions, then Ordinary Least Squares (OLS) regression produces unbiased estimates that have the smallest variance of all possible linear estimators. When these assumptions are violated, for instance, due to missing data introducing bias or inconsistency into the dataset, the estimates produced by the model can no longer be guaranteed to be optimal.

Missing Data Mechanisms

The theoretical foundation for understanding missing data was established by Rubin (1976) and later expanded upon by Little and Rubin. As described by the Statistical Consulting Centre of the University of Melbourne (n.d.), data is considered Missing Completely at Random (MCAR) if the probability of being missing does not depend on the values of any of the variables in the data, Missing at Random (MAR) if the probability of being missing depends only on observed data, and Missing Not at Random (MNAR) if the probability of missingness is related to the values that would have been observed had the data not been missing. Understanding which mechanism governs the missing values in a dataset is critical because it determines which imputation strategies are appropriate and what biases might be introduced if missing data is simply ignored.

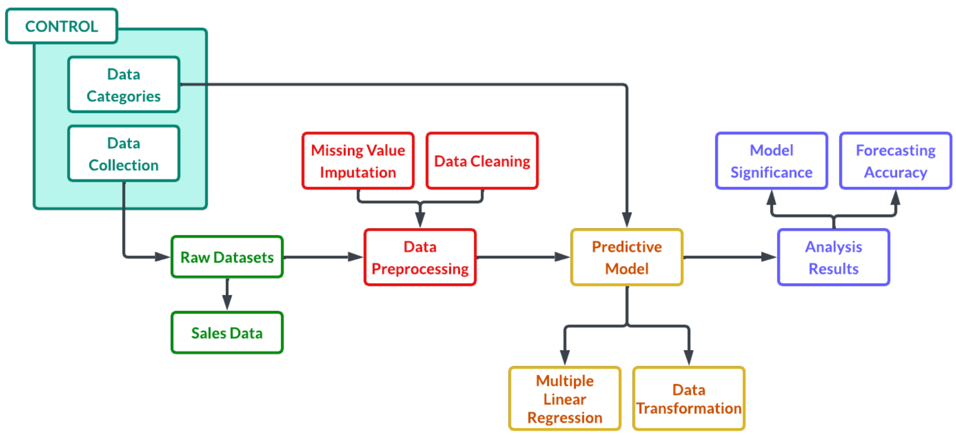
Further elaborated by APXML (n.d.), the MCAR assumption is the simplest scenario, where missingness is purely random and does not have a systematic cause related to the data, while MAR is often a reasonable working assumption when MCAR seems too simplistic and there is no strong domain reason to suspect MNAR. According to ScienceDirect (n.d.), the use of inappropriate methods for handling missing data can lead to bias in parameter estimates, bias in standard errors and test statistics, and inefficient use of data, all of which directly degrade the predictive accuracy of a regression model.

Data Preprocessing and Model Performance

Data preprocessing refers to a set of techniques applied to raw data before it is used for analysis or model training. As highlighted by Lee (2025), simple steps like normalization and feature scaling can have dramatic effects on model performance, and only including features that contribute significantly to prediction accuracy is essential to building a reliable regression model. In the context of sales prediction, preprocessing steps such as handling missing values, removing duplicate records, and encoding categorical variables are necessary to ensure that the dataset meets the assumptions required for valid regression analysis.

Supporting this, research cited in the IJEDR (2025) by Mehta and Sharma (2022) demonstrated that removing outliers and cleaning data improved model performance significantly, while Singh and Gupta (2022) compared various regression models and found that proper preprocessing influenced predictive results across model types. These findings collectively support the rationale of this study: that systematic application of data cleaning techniques prior to fitting a Multiple Linear Regression model will yield measurably improved performance in global sales prediction.

Conceptual Framework



The conceptual framework of this case study describes a flow of events detailing the different phases of data analysis it proposes.

At the start of the research process, the raw datasets, primarily consisting of sales data records, are aggregated into the Input Phase. The Data Collection procedures inherent to the source influence the initial integrity of this data, essential to determining the mechanism of missing records. Afterwards, the raw data undergoes Data Preprocessing. Missing Value Imputation (MVI) and Data Cleaning procedures are performed to eliminate and mitigate the impact of data loss and satisfy the assumptions of linear modeling.

These refined inputs are sent to the Predictive Model, utilizing the Multiple Linear Regression model and Data Transformation techniques which will be discussed further in the methodology of this paper. Data Categories defined in the dataset, such as platform and genre, are utilized as control variable, taken into account to ensure the model's validity and scrutinize market biases. Once all processes are finished, results of the analysis are printed, presented and evaluated through the lens of model significance and accuracy of variable forecasting, especially determining and confirming the reliability of the model in predicting global video game sales.

Chapter 3: Methodology

Research Design

Describe the design (e.g., descriptive, experimental).

Data Collection

The dataset utilized in this study was collected from Kaggle, named **Video Game Sales with Ratings** by Rush Kirubi from https://www.kaggle.com/datasets/rush4ratio/video-game-sales-with-ratings.

Table 1: First Six Rows of Video Game Sales Data Overview

Global_Sales	Platform	Game_Age	Genre
82.53	Wii	10	Sports
40.24	NES	31	Platform
35.52	Wii	8	Racing
32.77	Wii	7	Sports
31.37	GB	20	Role-Playing
30.26	GB	27	Puzzle

Critic_Score	Critic_Count	User_Score	User_Count	Rating
76	51	80	322	E
NA	NA	NA	NA	NA
82	73	83	709	E
80	73	80	192	E
NA	NA	NA	NA	NA
NA	NA	NA	NA	NA

Data Cleansing

The dataset have missing values **(NA)**. To address the missing values, the dataset was imputed using a **Predictive Mean Matching model**.

Table 2: First Six Rows of the Data After Imputation

Global_Sales	Platform	Game_Age	Genre
82.53	Wii	10	Sports
40.24	NES	31	Platform
35.52	Wii	8	Racing
32.77	Wii	7	Sports
31.37	GB	20	Role-Playing
30.26	GB	27	Puzzle

Critic_Score	Critic_Count	User_Score	User_Count	Rating
76	51	80	322	E
93	32	85	1588	E10+
82	73	83	709	E
80	73	80	192	E
80	66	92	709	E10+
92	96	85	594	T

Variables Description

Dependent Variable

- **Global_Sales**: We measure Global_Sales as the dependent variable, which is hypothesized to be influenced by the independent variables of platform popularity, game age, genre, critic scores and counts, user engagement, and content ratings.

Independent Variables

- **Platform**: May affect Global_Sales because some gaming platforms have larger user bases and stronger market demand than others.
- **Game_Age**: May influence Global_Sales since newer games often attract more buyers, while older games may experience declining sales over time.
- **Genre**: Can affect Global_Sales because certain game genres are more popular and appealing to players than others.
- **Critic_Score** May increase Global_Sales because games with higher review scores are more likely to attract consumers.
- **Critic_Count**: May affect Global_Sales since games reviewed by more critics usually gain greater visibility and public attention.
- **User_Score** May increase Global_Sales because games with higher review scores from users are more likely to attract more consumers.
- **User_Count**: May influence Global_Sales because higher player engagement and feedback often indicate stronger popularity.
- **Rating**: May affect Global_Sales by determining the target audience size and the accessibility of the game to consumers.

Data Analysis Techniques

- Descriptive Statistics
- Inferential Statistics

Chapter 4: Results and Discussion

Descriptive Statistics

Table 3: Descriptive Statistics of Dataset

	Global_Sales	Game_Age	Critic_Score	Critic_Count	User_Score	User_Count
X	Min. : 0.0100	Min. :-4.000	Min. :13.00	Min. : 3.00	Min. : 0.00	Min. : 4.0
X.1	1st Qu.: 0.0600	1st Qu.: 6.000	1st Qu.:60.00	1st Qu.: 12.00	1st Qu.:64.00	1st Qu.: 10.0
X.2	Median : 0.1700	Median : 9.000	Median :71.00	Median : 21.00	Median :75.00	Median : 24.0
X.3	Mean : 0.5335	Mean : 9.513	Mean :68.97	Mean : 26.36	Mean :71.25	Mean : 162.2
X.4	3rd Qu.: 0.4700	3rd Qu.:13.000	3rd Qu.:79.00	3rd Qu.: 36.00	3rd Qu.:82.00	3rd Qu.: 81.0
X.5	Max. :82.5300	Max. :36.000	Max. :98.00	Max. :113.00	Max. :97.00	Max. :10665.0
X.6	NA	NA's :269	NA's :8582	NA's :8582	NA's :9129	NA's :9129

	Platform	Genre	Rating
X	Length:16719	Length:16719	Length:16719
X.1	Class :character	Class :character	Class :character
X.2	Mode :character	Mode :character	Mode :character

Visualization

```r
library(ggplot2)
#ggplot(data, aes(x=Variable1, y=Variable2)) +
#  geom_point() +
#  theme_minimal()
```

Model

```r
#model <- lm(total_sales~., numeric_data)
#summary(model)
```

Interpretation

Discuss findings in paragraph form.

Chapter 5: Conclusion and Recommendations

Conclusion

Summarize key findings.

Recommendations

Provide practical suggestions

References

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Appendices

Appendix A: Code

```
# === R LIBRARIES ===

library(RKaggle)    # Access Kaggle
library(readr)      # File reading
library(dplyr)      # Data manipulation
library(tidyr)      # Data reshaping
library(stringr)    # String manipulation
library(forcats)    # Handling categorical factors
library(purrr)      # Functional programming
library(mice)       # Bayesian Linear Regression imputation
library(caret)      # Data splitting and pre-processing
library(car)        # Variance Inflation Factor (VIF)
library(lmtest)     # Testing for heteroscedasticity (Breusch-Pagan)
library(olsrr)      # Comprehensive OLS regression diagnostics
library(ggplot2)    # Main plotting engine
library(corrplot)   # Correlation matrices
library(knitr)      # Better summary table

# === DATA COLLECTION ===

# Get data from Kaggle
dataset <- read.csv("../Video_Games_Sales_as_at_22_Dec_2016.csv")

# Get data row count
data_row_count <- nrow(dataset)

# Drop all sales except global
dataset <- cbind(dataset[,-c(6:10)],dataset$Global_Sales)
dataset$Year_of_Release <- 2016 - dataset$Year_of_Release
names(dataset)[3] <- "Game_Age"
names(dataset)[12] <- "Global_Sales"

# Convert User Score to numeric
dataset <- dataset %>%
           mutate(User_Score = as.numeric(na_if(User_Score, "tbd")))
dataset$User_Score <- dataset$User_Score * 10

# Select only the columns you need by name
keep_columns <- c("Global_Sales",
                  "Platform",
                  "Game_Age",
                  "Genre",
                  "Critic_Score",
                  "Critic_Count",
                  "User_Score",
                  "User_Count",
                  "Rating")

# Overwrite the data with only those columns
```

```r
data_subset <- dataset[, keep_columns]

data_subset$Platform <- as.factor(data_subset$Platform)
data_subset$Genre <- as.factor(data_subset$Genre)
data_subset$Rating <- as.factor(data_subset$Rating)

# Show the table before imputation
knitr::kable(head(data_subset[, 1:4], 6),
             caption = "First Six Rows of Video Game Sales Data Overview")
knitr::kable(head(data_subset[, 5:9], 6))

# Predictive Mean Matching (PMM)
imputed_data <- mice(data_subset,
                     method = "pmm",
                     m = 1,
                     maxit = 5,
                     seed = 123,
                     printFlag = FALSE)

# Complete the dataset
data <- complete(imputed_data, 1)

# Show the table after imputation
knitr::kable(head(data[, 1:4], 6),
             caption = "First Six Rows of the Data After Imputation")
knitr::kable(head(data[, 5:9], 6))
```